Xi Zhang

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EDUCATION

- **Shanghai Jiao Tong University, *Shanghai, China***
  Ph.D., Department of Electronic Engineering
  - *Major in Digital Image/Video Processing* *Sep. 2015 – Sep. 2021*
- **McMaster University, *Hamilton, Canada***
  Visiting Ph.D., Department of Electrical & Computer Engineering
  - *Major in Image/Video Compression* *Oct. 2019 – Aug. 2020*
- **University of Electronic Science and Technology of China, *Chengdu, China***
  B.Sc., School of Mathematical Sciences
  - *Major in Mathematics and Physics Basic Science* *Sep. 2011 - Jun. 2015*

RESEARCH INTERESTS

- **Image/Video Compression and Restoration**
  Deep learning based image and video compression (e.g. lossy and lossless compression), restoration (e.g., compression artifacts reduction, super-resolution, low-light enhancement, denoising, etc.).
- **Deep Neural Networks**
  Generalization ability of deep neural networks, cognitive computing, brain-inspired network design.

SELECTED PUBLICATIONS

- **X. Zhang**, X. Wu. "Ultra High Fidelity Image Compression with l∞-constrained Encoding and Deep Decoding." IEEE Transactions on Image Processing (**TIP**), 2021.
- **X. Zhang**, X. Wu. "Multi-modality Deep Restoration of Compressed Face Videos." IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), Under Review.
- **X. Zhang**, X. Wu. "Attention-guided Image Compression by Deep Reconstruction of Compressive Sensed Saliency Skeleton." IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
- **X. Zhang**, X. Wu, X. Zhai, X. Ben, and C. Tu. "DAVD-Net: Deep Audio-aided Video Decompression of Talking Heads." *IEEE Conference on Computer Vision and Pattern Recognition **(CVPR**), 2020.*
- **X. Zhang**, X. Wu. "On Numerosity of Deep Neural Networks." *Advances in Neural Information Processing Systems (**NeurIPS**), 2020.*
- Y. Guo, **X. Zhang** and X. Wu. "Deep Multi-modality Soft-decoding of Vey Low Bit-rate Face Videos." *ACM International Conference on Multimedia **(ACM MM**), 2020.*
- X. Wu**, X. Zhang**, and Xiao Shu. "Cognitive Deficit of Deep Learning in Numerosity." *AAAI Conference on Artificial Intelligence (**AAAI**), 2019.*
- **X. Zhang**, X. Wu. "Nonlinear Prediction of Multidimensional Signals via Deep Regression with Applications to Image Coding." *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), 2019.*

- **X. Zhang** and X. Wu. "Near-lossless L-infinity constrained Image Decompression via Deep Neural *Network." Data Compression Conference (**DCC**). 2019.*
- **X. Zhang** and Xiaolin Wu. "Illumination invariant feature based on neighboring radiance ratio." *IEEE Visual Communications and Image Processing (**VCIP**). 2017.*

INTERNSHIP

- **Tencent**
  Algorithm Engineer Intern *Jun. 2019 – Sep. 2019*
  Media Algorithm Group, Video Research & Development Center, TEG.
  Researches on the restoration and enhancement of compressed videos.

SCHOLARSHIPS & AWARDS

- **SJTU Overseas Study Scholarship**
  SJTU, China *Oct. 2019*
- **Chinese National Scholarship**
  This scholarship is awarded to the top undergraduates in China. *Oct. 2014 & 2015*
- **The First Prize Scholarship**
  UESTC, China *Oct. 2013*
- **Outstanding Graduate of Sichuan Province**
  Sichuan Province, China *Jun. 2015*
- **Outstanding Graduate of UESTC**
  UESTC, China *Jun. 2015*
- **Outstanding Graduation Paper of UESTC**
  UESTC, China *Jun. 2015*
- **Second Prize in Chinese Contest of Mathematical Modeling**.
  Ministry of Education of the People's Republic of China *Dec. 2013*
- **Third Prize in National College Students' Contest of Mathematics.**
  Ministry of Education of the People's Republic of China *Dec. 2013*

SKILLS

- **Programming**: Python, Matlab, C/C++.
- **English level**: TOEFL 95, CET-6 530, CET-4 530.
- **Deep Learning Framework**: Pytorch, TensorFlow, Caffe.
- **Invited Reviewer**: NeurIPS, CVPR, AAAI, TIP